



TECHNICAL ASSIGNMENT

AI Research Intern

Binary Change Detection on EO-SAR Image Pairs

Position	Satellite AI Research Intern
Topic	Binary Change Detection — EO-SAR Image Pairs
Dataset	Data Link
Submission Deadline	13 May 2026, 11:59 PM IST
Submission Form	Form Link

1. OVERVIEW

This assignment is designed to evaluate how you approach an applied research problem end-to-end — from understanding the data to making design decisions, implementing a solution, and communicating your findings clearly.

The task is binary change detection on paired Electro-Optical (EO) and Synthetic Aperture Radar (SAR) imagery, a practically important problem in disaster response, urban monitoring, and environmental surveillance. You will be provided with the dataset along with a predefined train/validation/test split. Your goal is to build a model that classifies each pixel as changed or unchanged between a pre-event and post-event image pair.

Important: Use only the data provided. Do not use any external datasets or additional remote sensing imagery for training or fine-tuning.

We are not only evaluating your final metric scores. We are equally interested in how you think, what trade-offs you identify, and how clearly you communicate your reasoning. A well-reasoned approach with modest numbers is more valuable to us than strong numbers with no justification.

2. DATASET

2.1 *Provided Data*

The dataset contains co-registered EO-SAR pre/post-event image pairs with pixel-level annotation masks across multiple disaster events. The data will be provided with a fixed train/validation/test split — use this split exactly as provided without any modification.

Dataset Download: [DATA](#)

2.2 *Label Remapping (Mandatory)*

The original annotations contain four semantic classes. Before any training or evaluation, remap labels as follows:



Original Class	Original Value	Remapped Value	Remapped Class
Background	0	0	No-Change
Intact	1	0	No-Change
Damaged	2	1	Change
Destroyed	3	1	Change

This remapping must be applied consistently across all splits before any model training or metric computation.

2.3 Test Set — Partial Blind Evaluation

The test set provided to you contains only 50% of the full test data. You should report your metrics on this provided test split. The remaining 50% is held out at our end and will be used to evaluate your submitted model weights independently. Your model must generalise — optimising narrowly for the visible test split will reflect poorly in the blind evaluation.

2.4 A Note on the Data

Before writing any code, spend time understanding the dataset. Real-world remote sensing data has characteristics that significantly affect modelling decisions — class distribution, spatial resolution differences between EO and SAR modalities, noise profiles, and geographic diversity across events. Your report should reflect that you have looked at the data carefully before making any design choices.

3. TASK

Given a co-registered pre-event and post-event image pair (EO and SAR), your model must produce a binary pixel-level change mask:

- 1 = Change
- 0 = No-Change

You are free to use any AI-based approach. Architecture, training strategy, loss function, and use of pretrained weights are entirely your choice — but every significant decision must be justified in your report. Use of external datasets is not permitted.

Note on class imbalance: Change pixels are typically a small fraction of the total in disaster datasets. How you handle this is a meaningful design decision; we expect it to be addressed explicitly in your report.

4. EVALUATION METRICS

Report the following on both the validation split and the provided test split, computed for the change class (label = 1):

- Intersection over Union (IoU)
- Precision
- Recall
- F1 Score

Additionally, report a confusion matrix and comment on the error profile of your model — where does it fail, and why?

5. DELIVERABLES

Submit via the form linked in the metadata table. The form will require two items: (1) a public GitHub repository link, and (2) a ZIP file. Full submission instructions are in Section 7.

5.1 Code — GitHub Repository

Push all your code to a public GitHub repository and include the link in your submission. The repository must contain:



- All source code: dataloaders, model definition, training and evaluation scripts
- A configuration file (config.yaml or config.json) logging all hyperparameters: learning rate, batch size, epochs, optimizer, loss function, image size, augmentations, random seed, and any other relevant settings
- A well-structured README.md — see Section 5.1.1 below for required contents

The repository must be clean, well-organised, and fully reproducible. A reviewer should be able to clone it, follow the README, and reproduce reported metrics without needing to contact you.

5.1.1 README Requirements

The README is a critical deliverable. It must include all of the following:

Section	Content Required
Project Title & Description	Brief description of the task and your approach
Requirements	Python version, all dependencies with pinned versions (requirements.txt)
Environment Setup	Step-by-step conda/venv setup commands
Dataset Structure	Expected directory layout after placing the dataset
Training	<i>Command to run training from scratch, e.g. <code>python train.py --config config.yaml</code></i>
Evaluation	<i>Command to evaluate on test data by passing a path, e.g. <code>python eval.py --data_path /path/to/test --weights /path/to/checkpoint.pth</code></i>
Model Weights	Public link to download final checkpoint (Google Drive, HuggingFace Hub, etc.)
Results	Table of reported metrics on validation and test splits
Citation / References	Papers and codebases consulted

Model weights must be hosted on a public platform (Google Drive, Hugging Face Hub, or equivalent) and linked directly in the README — **do not submit weights as an email attachment or private link.**

5.2 Technical Report (PDF)

Your report must be crisp, precise, and well-structured. There is no page minimum or maximum — write as much as the work demands and no more. The report must include the following sections in order:

1. **Abstract.** A concise summary of the problem, your approach, and your key results. Should stand alone as a complete snapshot of your work.
2. **Literature Survey.** A thorough review of relevant work in change detection, EO-SAR fusion, and related areas. This is not a formality — engage seriously with the literature, identify the most relevant prior methods, and articulate what gap or direction your approach addresses. Cite all papers, codebases, and resources consulted.
3. **Methodology.** Your architecture, training strategy, loss function, handling of class imbalance, and the rationale behind each decision. If you experimented with multiple approaches, describe what you tried and why you moved on.
4. **Results.** IoU, Precision, Recall, F1, and confusion matrix on both the validation and test splits. Include qualitative visualisations — at minimum five prediction examples covering both success cases and failure cases.
5. **Future Work.** Assume you are joining GalaxEye as an intern and this assignment is your first-month deliverable. What would you do next? Be specific: what architectures, training strategies, or data strategies would you explore, and why?
6. **Conclusion.** Key takeaways and an honest assessment of the limitations of your current approach.

5.3 Time and Resource Log



Include as a final section in your report or as a separate plain-text file in the ZIP:

- Total time spent, broken down by activity: data exploration, literature reading, implementation, training, evaluation, and report writing
- Machine used for training: local or cloud (if cloud, specify provider and instance type), GPU model and VRAM, number of GPUs
- Approximate training time per epoch and total wall-clock training time
- Any resource or time constraints you worked under, and how they shaped your decisions

This log is not used to judge the quality of your work. It provides context useful when comparing submissions from candidates working under different resource constraints.

6. EVALUATION CRITERIA

Dimension	What We Look For
Data Understanding	Did you analyse the data before modelling? Did you identify its key characteristics and challenges?
Literature Survey	Is the review thorough and relevant? Do you demonstrate command of the state of the field?
Research Thinking	Are design decisions justified? Do you demonstrate awareness of trade-offs?
Implementation Quality	Is the code clean, reproducible, and well-organised?
Results and Analysis	Are metrics reported correctly on both splits? Is the error analysis insightful?
Future Work	Are proposed next steps specific, technically motivated, and feasible?
Communication	Is the report clear, concise, and technically precise?

7. HOW TO SUBMIT

Submit your completed work via the form linked in the metadata table on page 1. The form requires exactly two items — please ensure both are ready before opening the form.

7.1 *GitHub Repository Link*

Your repository must be set to public and contain all source code, a configuration file, and a README as specified in Section 5.1. The README must include a publicly accessible link to your model weights.

7.2 *ZIP File*

Submit a single ZIP file named `FirstName_LastName_GalaxEye.zip` containing:

- Model weights checkpoint file
- Technical report (PDF)
- Time and resource log (if not included in the report)
- Any other supporting documents referenced in your report

Important: Submissions with a private repository, missing model weights, or a corrupted ZIP will not be evaluated.



8. SUBMISSION NOTES

- Use only the dataset provided — no external remote sensing data for training or fine-tuning.
- Use of publicly available pretrained weights (e.g. ImageNet-pretrained backbones) is permitted — cite all external resources used.
- Use of LLMs or coding assistants for implementation is permitted.
- If you are unable to complete all sections due to time or compute constraints, submit what you have and include a note on what you would have done next.
- For queries, write to recruit_dil@galaxeye.space with subject: Assessment Query — [Your Full Name]. Queries received after 10 May 2026 may not receive a response before the deadline.

We look forward to reviewing your submission.

Good luck.